Nicole Jiang

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Education

University of Toronto 2024 – 2029

HBSc, Astronomy & Physics Specialist, Statistics Minor, Philosophy Minor

Relevant Coursework: Computational Astrophysics, Stars and Planets, Galaxies and Cosmology, Introduction to Computer Programming, Practical Physics I, Multivariable Calculus with Proofs, Thermal Physics, Probability and Statistics I

Technical Skills

Programming: Python (NumPy, SciPy, pandas, astropy, Matplotlib), TypeScript, SQL, Bash; Git; SSH

Data/ML: scikit-learn; PyTorch; Transformers

Web/Tools: React, HTML/CSS, Vite; LaTeX, Overleaf, Jupyter, VS Code, Microsoft Office, Google Workspace

Research & Technical Projects

Early Black-Hole Growth Constraints from JWST February 2026 – Present

Undergraduate Researcher, University of Toronto

- Built a provenance-tracked catalogue of 350 literature measurement records from 32 source families for accreting black holes and candidates at $z \geq 4$, preserving uncertainties, inference methods, assumptions, and quality flags
- Implemented vectorized NumPy growth diagnostics across seed masses, formation redshifts, Eddington ratios, spin-dependent efficiencies, and merger mass boosts
- Quantified probabilities that required lifetime-average Eddington ratios exceed one under fixed growth histories, using 10,000 Monte Carlo draws per object with reported mass errors and mass-scale shifts of up to ± 0.5 dex
- Identified 12 of 224 analysis objects requiring lifetime-average Eddington ratios above one for 100-solar-mass seeds formed at z=30, efficiency 0.1, and no merger boost; tested sensitivity to mass uncertainties and growth assumptions
- Led catalogue development and analysis under Prof. Pratika Dayal's mentorship (CITA, DAA-Dunlap)

Galaxy Star-Formation Main Sequence Analysis with Cosmological Simulations May 2025

- Queried 11,000,000+ galaxy records from public cosmological simulation databases (EAGLE, IllustrisTNG) and processed catalogues using Python, SSH sessions, and pickle serialization
- Visualized SFR–M* trends across simulation snapshots (redshifts $0 \leq z < 0.5$) using scatterplots and 2D histograms
- Compared single- vs multi-snapshot datasets to explore redshift-dependent trends in galaxy evolution
- Drafted an AAS-style scientific paper in LaTeX using the AAS journal submission Overleaf template

Predicting APA Site Choice from mRNA Sequences September 2025

- Runner-up in the Toronto Bioinformatics Hackathon as a five-member undergraduate team, placing 2/23 ($500 prize) out of 115+ individual contestants, including many graduate students, postdocs, and industry researchers
- Built the data preprocessing pipeline to clean and format raw RNA sequences for the DNABERT-2-117M transformer genome foundation model. Scaled the pipeline and batched inputs to train the model on a curated dataset of ~800,000 sequences
- Codified and enforced data-format standards for training; blocked non-conforming samples and implemented checks
- Co-developed project strategy and presented results to judges; model achieved 86% poly(A) site prediction accuracy

Saturn and Moons Observation with HDR Imaging October – December 2024

- Captured and compiled photos of Saturn and its moons, calibrating a pixel-to-kilometre conversion to measure distances
- Validated measurements by cross-referencing with 189 images taken over the span of two months, and comparing measurements over time to accurately evaluate Titan's orbital period

Service & Leadership

Journal Contributor September 2025 – Present

Royal Astronomical Society of Canada (RASC)

- Producing planet ephemerides for the Royal Astronomical Society of Canada's bimonthly Journal using Stellarium software
- Curating tables of noteworthy observational astronomy events for the Journal centrespread
- Also assisting with observatory maintenance on work trips to the E.C. Carr Astronomical Observatory

Student Host February – June 2024

Ontario Science Centre

- Designed and prototyped an interactive exhibition about the chemistry of art restoration, successfully pitching it to Science Centre staff to ultimately engage 100+ visitors with the final installation
- Created and presented a puppet show to elementary school classrooms across Toronto, teaching solar-eclipse concepts